SET – 5

1. Sarah is a botanist who studies different species of plants. She is particularly interested in the Iris genus and has collected data on the sepal length, sepal width, petal length, and petal width of various Iris flowers. She wants to use this data to classify the flowers into their respective species based on their physical characteristics. Sarah decides to use a machine learning algorithm for this task and trains a model on her collected data. The algorithm uses the sepal and petal measurements as input features and predicts the species of the flower based on these features. One day, Sarah is out in the field collecting new samples of Iris flowers. She measures the sepal and petal characteristics of each flower and inputs this information into the trained model. The model then predicts the species of each flower based on its physical characteristics.

   a) Read the IRIS.csv Data set using the Pandas module
   b) Plot the data using a scatter plot "sepal_width" versus "sepal_length" and color species.
   c) Split the data
   d) Fit the data to the model
   e) Predict the model with new test data [5, 3 , 1, .3]

2. Implement a Python program for the most specific hypothesis using Find-S algorithm for the following given dataset and show the output:

Example	Shape	Size	Color	Surface	Thickness	Target Concept
1	Circular	Large	Light	Smooth	Thick	Malignant (+)
2	Circular	Large	Light	Irregular	Thick	Malignant (+)
3	Oval	Large	Dark	Smooth	Thin	Benign (-)
4	Oval	Large	Light	Irregular	Thick	Malignant (+)

3. Develop a Python code for implementing Logistic regression and show its performance
4. Develop a Python code for implementing the Naive Bayes algorithm with and example

AIM & ALGORITHM (20)	PROGRAM (40)	DEBUGGING(10)	OUTPUT (20)	GITHUB UPLOAD (10)	TOTAL (100)

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SET-5

MLA0202 - Fundamentals of Machine Learning

Model lab exam

1. IRIS dataset classification

* **Aim**:

To read IRIS dataset, visualize sepal features, train a machine learning model & predict species of flower

* **Algorithm**

i) Import required libraries (Pandas, Matplotlib, Scikit learn).

ii) Load IRIS.csv using Pandas

iii) Plot sepal-width vs sepal-length with species.

iv) Train a classification model (Logistic Regression)

v) Predict species for new test data.

* **PROGRAM**

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
data = pd.read_csv("IRIS.csv")
x = data.iloc[:, 0:4]
y = data['species']
```

```
plt scatter (data ['sepal-length'], data ['sepal-width'],
  c= pd. factorize (y) [0])
plt.xlabel ("Sepal length")
plt. ylabel ("sepal width")
plt. show ()
model = logisticRegression (max_iter = 200)
model. fit (x_train, y_train)
prediction = model. predict ([[5, 3, 1, 0.3]])
print ("Predicted species", prediction)
```

* Output

Predicted Species : ['Iris-setosa']

* Result

the model successfully classified the Iris flower species

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2 Find S- Algorithm

* Aim :

to implement Find S--algorithm & determine the most specific hypothesis for given dataset

* Algorithm :

i) Initialize hypothesis with first +ve example.

ii) Compare each +ve example with hypothesis.

iii) Replace mismatched attributes with ?.

iv) Ignore -ve examples

v) Output the final hypothesis.

* Program

```
data = [ ['circular', 'large', 'light', 'smooth', 'thick', '+'],
         ['circular', 'large', 'light', 'irregular', 'thick', '+']
         ['oval', 'large', 'Dark', 'smooth', 'thin', '-']
       ]

hypothesis = None
for example in data:
    if example [-1]=='+' :
    if hypothesis is None :
    hypothesis = example [:-1]
    else:
  for i in range (len (hypothesis )):
  if hypothesis [i] != example [i]:
      hypothesis [i] = '?'
print (" Most specific hypothesis": hypothesis)
```

output

Most specific hypothesis : ['?', 'large', 'light', '?', 'thick']

* Result

the Find S- algorithm successfully generated most specific hypothesis consistent with +ve example.

3) Logistic Regression

* Aim : to implement logistic Regression & evaluate its classification performance using Iris dataset.

* Algorithm

i) Import required libraries.

ii) Split the dataset into training & testing of IRIS dataset

iii) Create Logistic Regression model

iv) Predict the output using test data

v) Calculate & display the accuracy of model.

* Program.

```
from sklearn.datasets import load_iris
from sklearn.linear_model import
    logisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
iris = load_iris()
x = iris.data
y = iris.target
```

```
model = LogisticRegression (max_iter = 200)
model.fit (x_train, y_train)
y_pred = model.predict (x_test)
print ("Accuracy", accuracy_score (y_test, y_pred))
```

* Output

Accuracy : 0.97

* Result

Logistic Regression was successfully implemented & achieved high accuracy in classification.

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4. Naive Bayes Algorithm

* Aim

to implement Naive Bayes algorithm for classifying data & evaluate its performance.

* Algorithm

i) Import required libraries
ii) load the iris dataset
iii) Create Gaussian Naive Bayes classifier.
iv) train model using training data
v) Predict the test data.

* Program

```
from sklearn.datasets import load_iris
from sklearn.naives_bayes import GaussianNB
from sklearn.metrics import accuracy_score.
iris = load iris()
x = iris_data
y = iris.target
model = GaussianNB()
modelfit(x_train, y_train)
y_pred = model.predict(x_test)
print("Accuracy", accuracy_score(y_test, y_pred))
```

* Output

Accuracy 0.95

* Result

Naive Bayes algorithm was successfully implemented.